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Marital Sorting: Explanations and Evidence

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Abstract and Keywords

This article reviews theoretical and empirical research related to marital sorting. The choice of marriage partner is a fundamental life decision that directly influences many individual and household outcomes that are central to individual well-being. Marital sorting is assumed to maximize the gains to marriage subject to the costs of marital search. Marital sorting, therefore, depends in part on the source of these gains. Mechanisms generating assortative mating, namely, preferences and search costs, are discussed in the context of sorting based on age, education, and race/ethnicity. The article explores the implication of marital sorting on income inequality and outlines future research avenues.

Keywords: marriage, marital sorting, marital search, assortative mating, age, education, race, ethnicity

Introduction

The choice of marriage partner is a fundamental life decision that directly influences many individual and household outcomes that are central to individual well-being. The resources available to the household are largely determined by the productive characteristics of the two spouses. Spouses must decide on a number of important joint choices, such as where to live, how many children to have and when, and how to invest in those children.

This chapter reviews topics and research related to marital sorting. To think about what type of person an individual will marry, it is necessary to first think about why people marry at all. The first section of the chapter discusses economic models of gains from marriage and their implications for marital sorting. The second section discusses economic models and research on marital sorting, with particular attention to the roles of preferences and search costs. The next three sections focus on marital sorting on three specific dimensions: age, education, and race/ethnicity. The final section discusses the interactions between marital sorting and income inequality, as well as the role of age of marriage.

Gains from Marriage and Marital Sorting

Individuals marry when utility in the married state exceeds utility in the single state. In other words, there must be gains from marriage. Marital sorting, or who marries whom, therefore, depends in part on the source of these gains.

Becker (1973, 1974, 1991) proposed gains through specialization as an important source of returns to marriage. Gains from specialization are generated when one spouse has a comparative advantage in market production and the other has a comparative advantage in household production. Crucially, in Becker's model, individuals have the choice to invest in capital that raises their productivity in the market or to invest in capital that raises their productivity in household production. Total output and gains from marriage are maximized when some individuals choose to invest only in household capital and then marry spouses who have chosen to invest only in market capital.

The principal role of marriage, therefore, in Becker's analysis, is to protect those who have invested entirely in household capital by ensuring the commitment of a spouse who will provide the household with market income. Without such commitments, it is not optimal for any individual to forego investment in market capital. With less specialization in human capital investment, gains from specialization, and therefore marriage, are diminished.

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The simplest marital sorting predicted by Becker's model is generated when men and women differ only in market wages. In this case, we would expect perfect negative assortative matching on wages. High-wage men should match with low-wage women and low-wage men should match with high-wage women. Within each household, the low-wage spouse spends more time in household production and the high-wage spouse spends more time in market production.

Becker does, however, predict that for many other traits, we should observe positive assortative matching, because these traits are likely complements in nonmarket production, particularly the production of children. He attributes the well-documented marital sorting based on traits such as education, religion, and ethnicity to complementarities in nonmarket production.

Economists have noted that technological changes in household production and rising wages of women have reduced gains from specialization as a primary source of gains from marriage. Researchers have therefore proposed that joint consumption of household public goods (such as children) and shared leisure are another source of marital surplus. In this case, larger gains to marriage are generated when individuals match with spouses with similar preferences for consumption (Lam 1988; Stevenson and Wolfers 2007; Lundberg 2012). For example, if children are household public goods, then investments one spouse makes in their care and well-being are nonexcludable and are enjoyed by the other spouse. Therefore, gains from marriage will be larger when spouses agree on which investments in their children are most beneficial. Similarly, one spouse's efforts to cultivate a beautiful flower garden in the backyard generate greater marital gains when he or she is partnered with a spouse who also derives great pleasure from backyard flower gardens.

As Lam (1988) points out, a limitation of Becker's approach to the economics of marriage is that his formal model does not include gains to marriage derived from joint consumption economies. Lam develops a model that includes a joint public good. He shows that gains from marriage from investment in a joint public good will generate positive assortative matching. While in Becker's formal model married couples only produce a single private commodity, Becker does acknowledge, in his discussion of assortative mating, that individuals have an incentive to marry spouses with similar preferences if similarities in consumption between spouses lower costs, or if some commodities are jointly consumed. In other words, it is beneficial for couples to agree on where they want to live, what sort of house they want to live in, how many children they want to have and when, and where they want to send them to school. Bergstrom (1997) refers to these as "public choices" that are made in the marriage.

Couples who generate marital surplus through specialization and exchange do not necessarily need to like each other or enjoy spending time together for the utility in the married state to dominate utility in the unmarried state. In contrast, couples who engage in relatively little specialization likely require compatibility of preferences and positive utility from joint consumption to generate sufficient marital surplus. Mansour and

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McKinnish (2014a) use the American Time Use Survey (ATUS) from 2003 to 2011 to test whether couples who engage in less specialization (e.g., are more similar in hours of market work) spend more time together. They find that among married couples without young children, those with a greater difference in weekly hours of work between husband and wife spend less time together on nonworking weekend days. Importantly, this relationship is quite symmetric between couples in which the husband works greater hours and couples in which the wife works greater hours. They do not find evidence of a relationship between specialization and couple time together among couples with young children. These results are consistent with a model in which couples who engage in less specialization experience gains from marriage due to joint consumption.

Lundberg (2012) provides additional analysis on the shift from gender specialization in marriage to joint consumption. She analyzes the effect of personality traits on selection into marriage using the German Socio-economic Panel Study and finds that among older cohorts, personality traits affect selection into marriage very differently for women and men, consistent with gender specialization in marriage. For example, agreeableness increases selection into marriage for women, but decreases it for men. This is consistent with selection into marriage of women who are nurturing and men with earnings power. In contrast, for younger cohorts, she finds no difference between men and women in how personality predicts marriage. This is consistent with marital surplus generated from joint consumption, rather than specialization. These results are therefore consistent with a shift over time from gains from marriage being generated by specialization to gains from marriage being generated by joint consumption.

The evidence in the literature, therefore, points to a shift over time toward greater realization of gains from marriage from joint consumption and fewer gains from specialization. We would therefore expect to see an increase over time in positive assortative matching, especially with regard to characteristics that predict investment in household public goods, particularly children. The observed increase in assortative matching on education, for example, is consistent with this prediction.

Models of Marital Sorting: Preferences and Search Costs

Assortative matching patterns in the population based on characteristics such as age and education likely reflect a preference by each of the spouses to maximize gains from marriage. Models of marital matching have focused on analyzing matching based on exactly one such characteristic and typically define marriage markets as broad and frictionless. In other words, they assume that all market participants have costless and full information about the conditions of the marriage market and all potential matches available to them. For instance, many studies have analyzed assortative matching patterns based on age (Choo and Siow 2006), income and education (Grossbard-

Schechtman 1993; Pencavel 1998), fertility (Chiappori and Oreffice 2008), or other types of preferences such as risk aversion (Legros and Newman 2007; Chiappori and Reny 2016).

Despite the many advantages of these models and their ability to explain certain patterns of assortative matching, they are not able to accommodate important features of matching that are observed in real life (Chiappori, Oreffice, and Quintana-Domeque 2012). First, the assumption of a frictionless and large marriage market characterized by different types might not be realistic. Recent empirical evidence, for example, suggests that the ease of meeting a potential spouse at college or work plays an important role in determining one's match (Kaufman, Messner, and Solis 2013; Mansour and McKinnish 2014b, 2016; Pestel 2016). Second, there is strong empirical evidence that spouses tend to be similar in a variety of characteristics such as age, religion, and appearance, suggesting that matching processes are not one-dimensional (Hitsch, Hortasçsu, and Ariely 2010; Chiappori et al. 2012).

The ability to consider the relative importance of different characteristics in the marriage market offers many advantages that allow researchers to address the limitations of more traditional matching models and to better understand the formation of different types of marriages. For example, previous research has noted that higher-weight women are more likely to marry lower-earning husbands and are less likely to be in a relationship (Averett and Korenman 1996; Averett, Sikora, and Argys 2008). Hamermesh and Biddle (1994) show a negative correlation between women's physical attractiveness and their husbands' education levels. These studies, however, rely exclusively on the weight and beauty measures as reported by women. As a result, the trade-offs they identify do not control for the men's corresponding physical attributes. Chiappori et al. (2012) overcome these issues by using data from the Panel Study of Income Dynamics (PSID), where reports of body mass indexes (BMIs) are available for both wives and husbands. Their findings suggest that men are willing to trade one extra year of education for up to two BMI units. Similarly, women are willing to accept a husband with 1.3 additional BMI units for a 1 percent increase in wages.

It is also useful to consider the implications of multidimensional matching in the presence of market frictions. For example, one could study matching processes when there is a shortage in the supply of men/women of certain attributes. Chiappori, Oreffice, and Quintana-Domeque (2014) study these issues in the context of matching between smoking and nonsmoking spouses. The main feature of their model is that the share of smoking men in the population is greater than the share of smoking women. Nonsmoking women prefer to match with nonsmoking men, but due to the shortage in the supply of nonsmoking men, their model predicts that nonsmoking women will "marry down," essentially accepting lower-educated nonsmoking men. The Chiappori et al. (2014) framework can be applied to other contexts in which individuals prefer to match within a certain category but face a shortage/surplus of prospective spouses within that category.

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Another mechanism that could explain patterns of assortative matching is related to search costs—that is, the ease or difficulty of finding a spouse with a given attribute (Kalmijn 1998; Kalmijn and Flap 2001). Using data from an online dating site, Hitsch et al. (2010) estimate matching preferences to predict stable matches under the assumption of frictionless search.¹ Although the assortative matching they generate is similar to patterns observed in the broader marriage market, they underpredict sorting by education and race/ethnicity, suggesting that meeting opportunities at school or work play an important role in marital formation. Belot and Francesconi (2013) use data from a speed-dating service in Britain to show that both men and women choose dating partners along an assortment of assortative traits, such as age and weight. Importantly, the nature of the speed-dating data allows them to analyze the impact of changing the set of potential dating partners on dating choices and conclude that meeting opportunities play a significant role in the observed matches. Using data from an online dating site in Korea, Lee (2009) finds that matches generated online exhibit less marital sorting by hometown or industry compared to the whole population, again indicating that search costs play an important role in the matching process.

In addition to evidence from online dating sites, a growing literature has documented the role of schools and workplaces as local marriage markets. For instance, McKinnish (2007) and Svarer (2007) found that individuals in sex-integrated workplaces have higher divorce rates, suggesting that individuals in marginal marriages face lower search costs in workplaces with many prospective mates of the opposite sex. Kaufman et al. (2013) use a regression discontinuity design and show that attending an elite college is associated with better-quality spouses for both men and women. Interestingly, they found the results to be more pronounced for men in fields with many women. Mansour and McKinnish (2014b) found that spouses with a large age difference are, on average, lower quality as measured by a host of characteristics, such as ability (as measured by the Armed Forces Qualification Test scores), education, and physical appearance. They provide evidence that low-quality individuals have age-diverse networks that facilitate marriages where the age gap between spouses is large. In contrast, high-skilled individuals have more homogenous age networks, especially during the part of the lifecycle in which marriages are likely to form.

The implications of the search cost mechanism suggest that early life choices such as the decision to go to school, what field to choose, and where to work can impact the matching process by changing the composition of people one interacts with most often, and not only through the marital surplus that such matches can generate. As a result, residential segregation, access to schooling, and job availability can directly impact marital formation. The increasing importance of such factors highlights the necessity to understand the relative significance of preferences and search cost mechanisms in the process of marital formation.

Mansour and McKinnish (2016) attempt to disentangle the relative importance of preferences and search costs by observing that married individuals are more likely to marry spouses who share their occupation, even if the share of perspective mates in one's

occupation is likely only a small fraction of the total number of available mates. Same-occupation matches could be more desirable if individuals have a preference to marry an individual who shares their occupation or if it is simply easier to find a spouse within one's occupation. Their empirical framework relies on the insights developed in Chiappori et al. (2014). If these matching patterns are generated because of a preference to marry similar-occupation spouses, then women should "marry down" in occupations where men are in short supply. Their findings, however, suggest the opposite, that women marry lower-quality husbands in occupations containing many men. This result is consistent with a model in which women are willing to trade off husbands' quality to avoid the search costs in the outside-occupation marriage market. They also show that these patterns are stronger in occupations that facilitate worker-to-worker interactions.

Sorting Based on Age

A well-documented feature of the marriage market is that individuals match assortatively based on age and that the most common pairing is one in which the husband is a few years older than the wife (Presser 1975; Glick and Lin 1986). This section first reviews the theoretical literature on marital age gaps and then discusses the empirical evidence.

Bergstrom and Bagnoli (1993) develop a model in which the delayed age of marriage for men, compared to women, and the resulting marital age gap, is explained by differences in household specialization between men and women. Central to their model is the assumption that men's value in the marriage market, meaning their earnings potential, is revealed at later ages than women's value in household production. All women marry young, because delays in marriage do not improve their productivity in home production and, therefore, their value on the marriage market. Higher-quality women (those with greater productivity in home production), however, marry higher-quality older men who have delayed marriage to reveal their high earnings potential. Lower-quality young women marry lower-quality young men, who do not bother to delay marriage because their earnings potential does not improve over time. In this model, both men and women in differently aged couples are higher quality than men and women in similarly aged couples.

Siow (1998) also develops a model that has the theoretical prediction that older men who marry younger women are financially successful. His model, like Bergstrom and Bagnoli's (1993), also has the feature that all women marry young (generated by declining fecundity). Young men all have the same wage, but some exogenously experience labor market success and have high wages as older men. Never-married and divorced old men are only able to marry or remarry young women if they are high wage.

Coles and Francesconi (2011) generate a similar prediction by assuming that both men and women receive utility from their partner's "fitness," which decays with age. Their model assumes that all workers earn low wages early in the lifecycle, but a fraction of

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workers receive high wages later in the lifecycle. If both men and women have similar probabilities of experiencing labor market success and receiving high wages at older ages, then we will observe both men and women who have experienced labor market success partnered with younger, fitter, but financially unsuccessful spouses.

While the previously discussed papers assume earnings differences across the lifecycle to generate pairings between older and younger spouses, Diaz-Gimenez and Giolito (2013) use only the fact that fecundity declines more rapidly for women than men over the lifecycle. Simply using these differences in fecundity, their theoretical model replicates key features of the US marriage market in terms of gender differences in age at marriage and marital age gap.

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Table 1 Distribution of Age Differences (%), US Men and Women, 1960–2000					
Age Difference	2000	1990	1980	1970	1960
	A. Full Sample				
+8 or more	9.79	9.92	9.05	10.58	12.76
+5 to 7	12.61	13.13	13.96	16.00	16.63
+2 to 4	29.35	31.49	33.98	33.71	31.54
+1 to -1	33.68	33.96	33.94	30.16	28.71
-2 to -4	8.96	7.27	6.23	6.37	6.85
-5 to -7	3.28	2.49	1.77	1.93	2.14
-8 or more	2.32	1.74	1.08	1.26	1.38
N	2,035,833	1,847,029	1,552,489	296,906	270,546
	B. Women with a College Degree				
+8 or more	8.15	8.37	7.06	8.47	9.09
+5 to 7	11.23	11.15	11.18	12.79	13.13

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+2 to 4	28.20	28.97	31.01	31.63	30.57
+1 to -1	39.42	40.54	42.51	38.77	35.94
-2 to -4	8.95	7.67	6.38	6.31	8.05
-5 to -7	2.63	2.11	1.29	1.45	2.16
-8 or more	1.43	1.18	0.57	0.58	1.05
N	555,444	388,500	236,776	27,368	16,685
C. Women without a College Degree					
+8 or more	10.41	10.33	9.40	10.79	13.00
+5 to 7	13.13	13.66	14.46	16.32	16.86
+2 to 4	29.79	32.16	34.52	33.92	31.61
+1 to -1	31.53	32.20	32.40	29.28	28.24
-2 to -4	8.97	7.16	6.20	6.38	6.77
-5 to -7	3.52	2.59	1.85	1.98	2.13

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-8 or more	2.65	1.89	1.17	1.33	1.40
N	1,480,389	1,458,529	1,315,713	269,538	253,861

Age difference is man's age minus woman's age.

Data is drawn from the 1960–2000 Integrated Public Use Microdata Series (IPUMS). Samples include couples aged 25 to 60 who are in their first marriage.

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Studying changes in the age difference between spouses can contribute to our understanding of the changing gains from marriage and offer insights into matching mechanisms. Table 1 includes descriptive statistics on the distribution of age differences using data from the 1960–2000 US census. The sample includes couples in their first marriage, aged 25 to 60, and the age difference is defined as the man’s age minus the woman’s age. Using the full sample in panel A, the share of similarly aged spouses has increased over time. For example, the proportion of couples where the man is no more than one year older or one year younger than his wife increased from about 29 percent in 1960 to about 34 percent in 2000. This increase seems to be driven by women with a college degree (defined as having four or more years of college education), whose share among married women increased from 6.2 percent in 1960 to 27.3 percent in 2000. For married women with a college degree, the share of couples with a similarly aged spouse (marital age gap of one year or less) increased from 36 percent in 1960 to about 40 percent in 2000 (panel B). A more modest increase in the share of similarly aged spouses appears to have occurred for women without a college degree, but the overall patterns still suggest that couples are becoming increasingly similar in age over this time period, with a peak observed in the 1990 census.

In contrast, the share of men who are five to seven years older than their wives decreased by about 4 percentage points over this time period. This decrease occurs for both women with and women without a college degree. Table 1 also shows that the share of women who are older than their husbands has increased over time, with the biggest increase concentrated among couples where the wife is two to four years older than her husband. This likely reflects the increase in the age at first marriage for women, but despite popular perceptions about older women marrying younger husbands, it is still relatively uncommon for women to be older than their husbands.

As we discussed, while theoretical models in economics generally predict that couples with a larger age gap should be positively selected on characteristics such as education and wage, recent empirical evidence suggests otherwise. Mansour and McKinnish (2014b) find that differently aged couples tend to be negatively selected. In National Longitudinal Survey of Youth 1979 cohort (NLSY79) data, men and women married to differently aged spouses have lower cognitive skills scores compared to those with similarly aged spouses. In US Decennial Census data, men and women with differently aged spouses have lower educational attainment and, conditional on educational attainment, work in lower-wage occupations and have lower annual earnings. Finally, Mansour and McKinnish (2014b) also analyze the National Longitudinal Study of Adolescent Health (Add Health) and find that men and women with differently aged spouses received lower ratings of physical appearance in high school.

Coles and Francesconi (2011) find that women who are older than their husbands are more likely to be economically successful than their husbands. Bloemen and Stancanelli (2015) find that married couples in which the husband is older by five or more years *or* the wife is older by five or more years are both more likely to be “wife sole provider”

couples than similarly aged couples. The results in Mansour and McKinnish (2014b) suggest that these previous findings reflect the fact that men with differently aged spouses on average have much lower earnings outcomes.

Much of the theoretical literature assumes that, conditional on income, individuals receive higher utility from younger, more fecund or more attractive, partners. Other research suggests that individuals might receive utility from similarly aged spouses. For example, Choo and Siow (2006) develop and estimate a structural model of the marriage market. They estimate systematic net gains to marriage based on age of husband and wife, and find that women maximize their net gain with slightly older husbands and men maximize their net gain with slightly younger wives. They also find that the distribution of net gains by partner's age becomes more dispersed as an individual ages. These gains are inferred from the observed matching rates. In other words, because the most common pairing is one in which the husband is slightly older than the wife, the model attributes the greatest net gains to such a match.

Analysis of online and speed-dating contexts indicates that both men and women prefer to contact similarly aged prospective mates (Belot and Francesconi 2013; Hitsch et al. 2010). To the extent that joint consumption and similarity in preferences are increasingly important sources of gains from marriage, this would seem to make similarly aged pairings more attractive. Spouses who are at similar points in their lifecycle are more likely to agree on joint choices such as location, housing, children, and leisure activities. In fact, there is evidence that the age difference between spouses is negatively related to marital stability and life satisfaction (Cherlin 1977; Lillard, Brien, and Waite 1995; Lee and McKinnish 2016).

For instance, Lee and McKinnish (2016) analyze whether married individuals are more satisfied with similarly aged spouses or differently aged spouses using the 2001–2013 Household, Income, and Labor Dynamics in Australia (HILDA) data. They find that both men and women on average report greater satisfaction with younger spouses and less satisfaction with older spouses. For both men and women, marital satisfaction declines more rapidly over the duration of the marriage for those married to differently aged partners compared to those married to similarly aged partners. These steeper declines erase the initial higher levels of marital satisfaction experienced by men married to younger wives and women married to younger husbands.

Sorting Based on Education

One of the most studied dimensions of assortative mating has been the tendency for individuals with similar educational attainment, especially those with college degrees, to marry one another (Mare 1991; Kalmijn 1998; Schwartz and Mare 2005). In contrast to assortative mating based on ethnicity, race, or socioeconomic background, education is an acquired trait, a choice that an individual makes early in adulthood (Mare 1991). As

such, educational attainment is influenced by a person's background, the chances of labor market success, and the potential benefits from education in the marriage market. In turn, education influences spouse choice and quality, the source of gains from marriage, and other key elements such as family size and investments in children.

Patterns of educational assortative mating are especially pronounced at higher levels of education and have been increasing over time (Mare 1991; Blossfeld 2009). They also coincide with the rapid rise of female college enrollment relative to the enrollment of men, a dramatic change that women in many countries have experienced during the second half of the twentieth century (Goldin, Katz, and Kuziemko 2006).² Thus, discussing the sources of the rise in educational assortative matching has to account for changes in women's education and their increased role in the labor market relative to home production.

Generally, researchers have decomposed returns to schooling into labor market returns and nonpecuniary returns, which among them is the marriage market return (Goldin 1992; Oreopoulos and Salvanes 2011). The increase in the returns to schooling for women can be attributed, in part, to lower discrimination against women in the labor market, improvements in the technology of home production, and changes in the norms about the role of women in the household (Chiappori, Iyigun, and Weiss 2009; Albanesi and Olivetti 2009). Higher labor market returns to education enable college-educated women to increase the possibility of higher household income by marrying similarly educated men, while also changing their relative bargaining power within the marriage and improving their outside options (Lundberg and Pollak 1993; Iyigun and Walsh 2007).

Marriage market conditions also contribute to the degree of educational assortative mating and to the premarital decisions to invest in schooling. One important determinant of the degree of assortative mating is the ratio of men to women in the population, that is, the sex ratio. Theoretical models predict that a decrease in the sex ratio improves the outcomes of men in the marriage market and worsens those for women (Becker 1973; Bloch and Ryder 2000; Chiappori, Fortin, and Lacroix 2002). Empirically, Abramitzky, Delavande, and Vasconcelos (2011) analyze how male scarcity in post-World War II France impacted assortative matching and show that a decrease in the sex ratio allowed men to marry women from higher social class as measured by men's occupations. Charles and Luoh (2010) study the impact of changes in the sex ratio due to the increase in the male incarceration rate in the United States. They find that a decrease in the availability of marriageable men decreases the likelihood of women to ever marry, and for those who do marry, they find an increase in the likelihood of matches where the woman is more educated than her spouse.

Interestingly, however, Charles and Luoh (2010) also found that women in marriage markets mostly affected by male incarceration have increased their schooling and labor supply in response to the decrease in the sex ratio. These patterns suggest that premarital investments in schooling are likely endogenous to marriage market conditions. Iyigun and Walsh (2007) and Chiappori et al. (2009) formalize this idea by introducing a

marriage market in which the returns to schooling for women exceed those for men because of the higher marriage market return for women. They show that under such conditions, the share of educated women will exceed that of men, leading to a change in the marital surplus and the way it is divided. The excess supply of educated women implies that some educated women will marry uneducated husbands but that uneducated women marry only uneducated husbands.

Lafortune (2013) provides empirical evidence on the relationship between the sex ratio and premarital investments in schooling. Following Angrist (2002), she exploits the imbalance in the sex ratio faced by second-generation Americans in their state and ethnic group at the turn of the twentieth century. She finds that an increase in the share of men to women in a given state and ethnic group induces men to obtain more education. In contrast, females from ethnic groups that have high rates of intermarriage choose to acquire less formal schooling. These results suggest that marriage market conditions have a long-run effect impacting not only the likelihood of marriages and matching but also investments in schooling.

The increase in the educational attainment of women in recent decades has also contributed to the rise of higher education as an important marriage market (Blossfeld 2009). In most countries, individuals attend college around the same age when they are searching for a mate, creating peer groups that are homogenous in both age and expected educational attainment (Blossfeld and Timm 2003). As a result, interactions in college, through networks of friends, or in posteducation employment settings increase the chances of meeting a spouse with a similar educational background and contributes directly to the formation of homogamous marriages where both spouses share many attributes such as age and educational attainment (Blossfeld and Timm 2003; Mansour and McKinnish 2016).

Although sociologists have pointed out the importance of educational systems as marriage markets, this mechanism has received less attention from economists, despite some recent evidence on the relationship between attending elite universities and the choice of field of study on a person's marital outcomes (Kaufman et al. 2013; Pestel 2016). For example, Kaufman et al. (2013) use a regression discontinuity design to study the effects of attending an elite university in Chile on marriage outcomes. They find that women who were just above the admission cutoff to attend a higher-ranked university marry higher-quality men compared to similar women who happen to be just below the admission cutoff and who had to attend a slightly lower-ranked college. Pestel (2016) takes advantage of the variation in sex ratios within fields of study and over time and finds that women in fields where men are scarce marry lower-quality husbands. Similarly, he finds that in fields where men are in excess supply, they are more likely to "marry down" compared to fields where men are scarce. Mansour and McKinnish (2016) present empirical evidence that much of the observed spousal matching based on field of study among college graduates is driven by the lower search costs of meeting potential

partners in your own field of study, rather than preferences for partners from the same field of study.

Sorting Based on Race and Ethnicity

In addition to sorting based on characteristics such as education and age, individuals disproportionately marry within racial/ethnic category. This pattern of racial/ethnic sorting, or endogamous marriage, can be generated by preferences for racial/ethnic similarity with a spouse. Spouses from the same ethnic background may, for example, have more similar preferences for joint consumption of cultural and/or religious practices, particularly with regard to raising and investing in children. Marital sorting along racial/ethnic lines can also be generated if there are “taboos” or perceived social costs of marrying an individual from an ethnic category with lower average social economic status, even conditional on the individual’s own productive characteristics. In this case, the individual from the category that generates these costs would have to compensate a spouse who engages in intermarriage. Under these circumstances, as long as there are similar-quality prospective mates from their own racial/ethnic category who would not require such compensation, intermarriage will be rare. Additionally, such sorting could result from differences in meeting opportunities if individuals are more likely to live, work, go to school, and attend places of worship where they meet a disproportionate number of prospective mates from their own racial/ethnic group.

Furtado and Theodoropolous (2011) analyze the relationship between ethnic and educational similarities and find that individuals with higher levels of schooling are more likely to match across ethnicities. The authors interpret this schooling effect as evidence that education facilitates the ability to adapt to different customs, as well as the communication and negotiation skills to navigate differences between spouses in cultural backgrounds and expectations. They also find that for native-born individuals and immigrants who arrived at young ages, higher education increases the odds of endogamous matching for those who belong to a high-education ethnicity, but not for those who belong to a low-education ethnicity. They argue that this suggests that while immigrants who arrive as teenagers have a preference for matching based on ethnicity, native-born and early-arriving immigrants are more concerned with matching based on education. Finally, they find no evidence that education affects intermarriage of Asians, which indicates strong preferences for endogamy in this group.

Qian (1997) also finds evidence of strong preferences for endogamous matching among Asians. While Asian Americans have a high rate of intermarriage, this is largely due to the fact that they are a relatively small fraction of the population, so that only a small fraction of their prospective partners in the marriage market are also Asian. Once group size is taken into account, Asians are more endogamous than whites or Hispanics, but less endogamous than African Americans. There is also evidence that Asian women who marry

white husbands “marry up” in the husband’s education compared to Asian women who marry Asian husbands.

Wong (2003) investigates the low rate of intermarriage between black men and white women by estimating a structural model that allows for a “mating taboo.” She finds that the taboo, or preferences, explains the majority of the shortfall in this form of intermarriage. Gullickson (2006) shows that higher-educated blacks are more likely to intermarry and that white women–black men pairings are more educationally hypergamous, meaning that wives in these pairings are disproportionately likely to be marrying up in terms of the spouse’s educational attainment. These results are consistent with a social cost to intermarriage that must be compensated for the white spouse. Also consistent with this idea of compensation for intermarriage are the findings from Grossbard, Gimenez-Nadal, and Molina (2014) on household work and intermarriage. Using the American Time Use Survey for 2003–2009, they find that both white men and women on average perform less household work when married to a black spouse compared to those married to white spouses. Similarly, they find that black men and women on average perform more household work when married to a white spouse than when married to black spouses.

Angrist (2002) studies marriage among second-generation immigrants in the United States using US census data from 1910 to 1940. During this period, rates of endogamous matches among native-born children of immigrants were still quite high. Focusing on ten ethnic groups (e.g., Italian, Irish, British, Russian/Polish, Mexican), he makes use of the fact that within some ethnic groups, the sex ratio (number of men to women) is much more favorable for women wanting an endogamous match than others. He finds that second-generation women had higher marriage rates, obtained higher-quality husbands, and had lower levels of labor supply when the within-ethnicity sex ratio was more favorable.

Most of the papers on ethnic and racial intermarriage focus on the role of preferences in generating marital sorting by ethnicity, but it is likely that meeting opportunities also play a role. For example, the higher rates of intermarriage for more highly educated black and Hispanic individuals documented by Furtado and Theodoropolous (2011) could in part reflect the fact that more blacks and Hispanics with high education attainment search for partners in more integrated school and work environments. An intriguing piece of suggestive evidence on the role of meeting opportunities and intermarriage is found among those with military service. Fryer (2007) discusses the fact that while intermarriage rates were similar among veterans and nonveterans up until 1960, since then intermarriage rates have increased faster for white veterans than white nonveterans. It is not possible from the raw statistics to distinguish between competing explanations of selection into the armed forces of individuals more prone to intermarriage, an effect of military service on preferences for intermarriage, or whether military service facilitates marital search in a more racially and ethnically heterogeneous

pool of potential spouses. Fryer does point out that the rise in intermarriage among veterans corresponds to the timing of integration of the armed forces.

Inequality, Age of Marriage, and Marital Sorting

Marriage market researchers have pointed out potential interactions between marital sorting and income inequality (Schwartz 2013; Autor 2014). Increased income inequality can generate a greater degree of sorting based on socioeconomic status. If, for example, returns to education increase, so that differences in average income across education groups also increase, then it becomes more costly for a high-education individual to marry a spouse with low educational attainment (Fernández, Guner, and Knowles 2005; Schwartz 2013). The costs to “marrying down” in education are even greater if the increased income inequality is accompanied by increased residential sorting and growing disparities across neighborhoods in school quality. This suggests that as economic inequality increases, we should see increased positive assortative matching based on education.

Fernández et al. (2005) analyze household-level surveys from thirty-four countries using data from the Luxembourg Income Study (LIS) and the Inter-American Development Bank (IDB). They find a significant positive relationship between returns to education and marital sorting based on education. Carbone and Cahn (2014) argue that growing earnings inequality combined with growing gender equality explains the observed rise in positive assortative matching in the United States that started in the 1970s. Cornelson and Siow (2016) point out that Carbone and Cahn (2014) only use stylized facts and qualitative evidence to support their argument, and that while the correlation between returns to education and marital sorting has been documented, no one had previously assessed how much of the increased positive assortative matching could be explained by increased inequality. Cornelson and Siow (2016) therefore empirically evaluate the Carbone and Cahn claim using state-level data from the decennial censuses and American Community Surveys (ACSs) for the years 1970–2012. They find that while there is a positive relationship between wage inequality and sorting based on education, the relationship is not strong enough to explain much of the observed changes in marital sorting during this time period.

It is also the case that increased marital sorting can exacerbate income inequality. If high-earning men marry high-earning women and low-earning men marry low-earning women, this will magnify individual-level earnings inequality, resulting in higher overall household-level income inequality. The effect of sorting by education on income inequality will be diminished to the extent that women’s labor force participation is low or responds negatively to husbands’ earnings. Greenwood et al. (2014) use census data from 1960 and 2005 to calculate how much the increase in positive assortative matching contributed to

the concurrent rise in household income inequality. They find that imposing random matching in the 2005 data only has a modest effect on the calculated Gini coefficient. The rise in single-person households, due to a decline in marriage and rise in divorce, is a much more important explanation for the rise in household-level inequality during this period.

Goldin and Katz (2002) investigate how the increased availability of oral contraceptives to college-aged women in the 1960s and 1970s changed their marriage and career investment behavior. Their model predicts that by lowering the cost of marital delay, high-ability women will delay marriage and invest in human capital and careers. This allows them to realize their full potential in the labor market, making them more attractive to high-ability men. The Goldin and Katz model therefore predicts that changes in conditions that encourage women to delay marriage will also lead to greater assortative matching. Carbone and Cahn (2014) argue that the increase in educational attainment of women allowed high-education women to differentiate themselves on the marriage market from low-education women, encouraging positive assortative matching. The result, they argue, is that low-education women are then left behind with only low-education men available as marriage partners, who themselves were experiencing declining labor market outcomes.

Loughran (2002) argues that higher male wage inequality increases the returns to marital search, and therefore should generate delays in marriage for women. Using data on 124 Standard Metropolitan Statistical Areas (SMSAs) from the 1970, 1980, and 1990 US census data, he finds that rising wage inequality explains 7 to 18 percent of the decline in marriage rates for white women aged 22 to 30. He does not predict or test for changes in assortative matching. Similarly, Gould and Paserman (2003) find that women living in cities with high income inequality spend more time searching for a husband. According to their estimates, rising income inequality explains about 25 percent of the decline in marriage rates in the past few decades.

While models such as Goldin and Katz (2002) predict that marital delay will increase assortative matching, it is not the case empirically that later age of marriage predicts more homogamous matching. This is because the marriage pool changes as an individual ages. To illustrate this with a simple example, suppose that 35 percent of men and 40 percent of women in their mid-20s are college graduates. Assume that there is very strong assortative matching so that almost all of the college-educated men marry college-educated women. It is easy to see that the college-educated women who did not marry sufficiently early in the lifecycle will end up eventually with a marriage market in which unmarried college-educated women substantially outnumber the remaining unmarried college-educated men. Those women who have delayed marriage will increasingly have to marry down in the spouse's education to match with a spouse, but the same is not true for men. Akin and Platt (2016) develop and calibrate a model that documents that spousal quality (as measured by educational attainment) declines more rapidly with age at marriage for women than men. Results from their calibration of a nonstationary

sequential search model indicate that, as women age, they drop their reservation value for spouse quality more rapidly than men due to the dwindling pool of educated men and oversupply of educated women.

Conclusion

Changes in the gains from marriage and the ways in which people meet prospective spouses are likely to change the degree of marital sorting. Whether these changes will decrease or increase the degree of assortative mating based on characteristics such as age, education, or race/ethnicity depends on the relative importance of preferences versus search costs.

For instance, if current assortative matching based on age and race is due, in part, to the homogenous composition of search pools, then the rise of Internet dating should reduce the cost of searching in more diverse pools. In such a case, the likelihood of interacting with prospective partners with diverse demographic characteristics will lead to a decrease in the degree of sorting based on attributes such as education, age, and race, potentially reducing household-level inequality.

On the other hand, lower search costs can increase sorting based on dimensions about which individuals have strong preferences, but for which sorting is limited by thin markets. For example, if some individuals have a strong preference to match with someone who shares their religion or ethnicity, then online dating may increase sorting based on these characteristics. The extent of the increase in sorting in this scenario will be larger for individuals from small religious or ethnic groups who face thin marriage markets. It is also important to note that the interaction between lower search costs, thin markets, and preferences could have bigger impacts at the tail of the distribution of a given characteristic. For instance, lower search costs would allow the very most physically attractive individuals to find individuals who are similar to them in attractiveness, leading to an increase in sorting based on beauty at the tail of the distribution.

Disentangling the importance of preferences and search costs will crucially depend on the development of new theoretical models that can accommodate the possibility of heterogeneous search costs in addition to frictions such as the sex ratio. Such models, combined with the availability of new data sets on how individuals meet their spouses, are guaranteed to improve our understanding on how modern marriages form and on their wider impacts on society.

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Notes:

⁽¹⁾ A match is defined as stable if it is not possible to find a man and a woman who are willing to abandon their current partners and match with each other (Hitsch et al. 2010).

⁽²⁾ Hoffman and Averett (2016) report that in 1970, the share of college graduates among US men and women ages 25 to 34 was 19.7 percent and 12 percent, respectively. By 2014, the share of female college graduates in the same age group has increased to 39 percent, while the share of male college graduates has increased to 31.6 percent.

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